Do Huu Dat

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Website | Scholar | Github

EDUCATION

VinUniversity — Oct 2022 – Jun 2026
Bachelor of Science., Major: Computer Science — Hanoi, Vietnam

POSTECH — Feb 2024 – Jun 2024
Exchange Student — Pohang, Korea

EXPERIENCE

Research Intern — June. 2025 – Now
CogAI4Sci @ NUS (PI: Prof. Dianbo Liu) — Singapore

- Leading a project investigating emergent compositionality in neural networks, exploring when and how gradient descent enables MLPs to form unified, factored, meaningful representations.

Research Intern — Feb. 2024 – Nov. 2024
Algorithmic Machine Intelligence Lab @ KAIST (PI: Prof. Taehyun Oh) — Pohang, Korea

- Led the project **VSC**, addressing the failure of diffusion models to bind multiple attributes to corresponding objects correctly. Proposed a compositional framework that decomposes prompts into sub-prompts, generates sub-images, and fuses their embeddings, achieving stronger semantic alignment and robust out-of-distribution generation.
- Leading a project on novel-view 3D object reconstruction, tackling the lack of structural consistency in diffusion-based synthesis. Introduced symmetry-based priors and voxel-level equivariance constraints for test-time optimization, leading to significant improvements in geometric fidelity and multi-view consistency.

Undergraduate Researcher — Oct. 2022 – June. 2025
VinUniversity (PI: Prof. Wray Buntine, Prof. Laurent El Ghaoui) — Hanoi, Vietnam

- Led the development of **HOPE**, a framework to address the limited compositional generalization of pre-trained models like CLIP. Proposed a memory-based primitive retrieval system, enabling recognition of unseen concept combinations with improved performance.
- Co-led the project tackling the model collapse phenomenon in long-sequence generation of the discrete diffusion model. Introduced a semantic-aware noising process that perturbs less important tokens first, and CrossMamba, an encoder–decoder variant aligned with random noise. achieving faster inference and stronger semantic coherence than prior baselines. This project has been accepted to **NAACL 2025**.
- Developed implicit root-finding computation of Transformer and Mamba. Benchmarked the convergence and stability of ADMM, projected gradient descent, and Levenberg-Marquardt algorithms in solving the large-scale root-finding.

Machine Learning Engineer — Oct. 2023 – Feb. 2024
Upwork — Remote

- Implemented research methodologies from academic papers, focusing on Transformers for multi-agent reinforcement learning using PyTorch.

PUBLICATION

(*Equal contribution are denoted by "*".*)

Conference

[C2] **Do Huu Dat**, Nam Hyeon-Woo, Po Yuan Mao, Tae-Hyun Oh, "VSC: Visual Search Compositional Text-to-Image Diffusion Model", ICCV 2025, pdf

[C1] **Do Huu Dat***, Do Duc Anh*, Luu Anh Tuan, Wray Buntine, "Discrete Diffusion Language Model for Efficient Text Summarization", NAACL 2025, pdf

[C0] **Do Huu Dat**, Po Yuan Mao, Nguyen Tien Hoang, Wray Buntine, Mohammed Bennamoun, "HOPE: A Memory-Based and Composition-Aware Framework for Zero-Shot Learning with Hopfield Network and Soft Mixture of Experts", WACV 2025 **oral**, pdf

PROJECTS

CLIP-NAV: Pretraining Image-Language for Drone Navigation github — Apr. 2022 – Nov. 2022

- Leveraging the CLIP model's zero-shot learning abilities to address limitations in visual navigation and zero-shot semantic segmentation for controlling drones with textual descriptions.

Re-implementing SODA github — Nov. 2023 – Dec. 2023

- Implementing the paper "SODA: Bottleneck Diffusion Models for Representation Learning" in PyTorch.